Feier (Fay) Lyu

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RESEARCH INTERESTS

Optimization: First-order methods, nonconvex and minimax optimization, optimization landscapes and stationary-point geometry, negative curvature, graph-structured optimization.

Formal Methods for Optimization: Isabelle/HOL, mechanized convergence and approximation proofs, proof engineering.

EDUCATION

University of Edinburgh Edinburgh, UK
BSc in Mathematics 09/2023 – 07/2027 (Expected)

- Performance: Average (Years 1–2): 78/100 (First-Class range).
- Key Grades: Linear Algebra (97%), Probability (86%), Algebra & Calculus (87%).

Cornell University Ithaca, NY, USA
Undergraduate Exchange Program, College of Arts & Sciences 08/2025 – 05/2026

- Coursework Focus: Operations Research and Information Engineering (ORIE) and Mathematics; GPA: 3.723/4.3.
- Selected Grades: Undergraduate Research in ORIE (A+), Optimization for AI (A), Linear Algebra (A), Optimization I (A−), Introduction to Game Theory (A−).

FORMAL PROOF PUBLICATIONS

First-Order Methods for Smooth Convex Optimization in Isabelle/HOL
Feier Lyu. *Archive of Formal Proofs*, July 2026. [AFP Entry]

- Independently developed reusable Isabelle/HOL infrastructure for smooth first-order optimization; formalized projected gradient descent with geometric and analytic interfaces for descent lemmas, residual certificates, and telescoping bounds.

Greedy Algorithms for Cardinality-Constrained Submodular Maximization
Feier Lyu. *Archive of Formal Proofs*, May 2026. [AFP Entry]

- Mechanized the classical greedy approximation guarantee for monotone submodular maximization under a cardinality constraint, verifying the approximation ratio $1 - (1 - 1/k)^k \geq 1 - 1/e$, together with a verified stateful lazy-greedy variant using cached marginal-gain bounds.

RESEARCH EXPERIENCE

University of Edinburgh Edinburgh, UK
Undergraduate Honours Project (Supervisor: Dr. Akshay Gupte) 10/2026 – Present
Root-Finding Algorithms for Graph Edge Expansion

- Implemented an exact-arithmetic evaluator for the parametric edge-expansion problem, together with Newton–Dinkelbach, fixed-point iteration, and Steffensen's method.
- Derived a contraction bound for the fixed-point map and validated root accuracy and stopping conditions against independent enumeration on ten test graphs.

RESEARCH EXPERIENCE (CONTINUED)

Cornell University (ORIE Department) Ithaca, NY, USA
Undergraduate Researcher (Advisor: Prof. Shoham Sabach) 01/2026 – Present
Collaborator: Owen Li
Nonconvex Network Localization: Stationary-Point Geometry and Negative Curvature

- Jointly derived a Dirichlet-energy condition whose threshold is expressed through effective resistance, sharpening negative-curvature certificates based on vertex degree and minimum cut.
- Independently constructed a regular-pentagon stationary family exhibiting a parameter range where negative curvature is undetected by the single-edge effective-resistance, minimum-cut, and degree certificates studied; derived its full planar Hessian spectrum via Fourier-mode decomposition.
- Independently derived an ambient-dimension Hessian block decomposition, exhibiting a strict local minimum modulo rigid motions in $\mathbb{R}^2$ that becomes a strict saddle after zero-padded lifting to $\mathbb{R}^3$.

University of Edinburgh Edinburgh, UK
Undergraduate Researcher (Advisor: Dr. Wenda Li) 06/2025 – Present

- Under Dr. Li's supervision, developed an Isabelle/HOL formalization of greedy and lazy-greedy algorithms for cardinality-constrained monotone submodular maximization, resulting in a sole-authored *Archive of Formal Proofs* entry.
- Exploring a post-publication extension toward stochastic greedy methods and executable formalization.

INDUSTRY EXPERIENCE

Liblib AI 06/2025 – 07/2025
Data Engineering Intern

- Wrote SQL queries in Alibaba Cloud DataWorks to handle complex data requests from operations teams, including multi-source table joins, window aggregations, filtering pipelines, and result consistency validation.

PATENTS

Chinese Invention Patent Application: Anti-Segregation Spreading Method and Device
Application No. 2024112756051, pending.
Proposed a mechanical framework to enhance material homogeneity in industrial spreading processes.

Chinese Utility Model Patent: Anti-Segregation Spreading Device
Patent No. ZL202422234130.3, authorized 4 July 2025.
Implementation of a physical distribution system for industrial material consistency control.

SKILLS & INTERESTS

Technical Skills: Isabelle/HOL, Python (NumPy, SciPy), LaTeX, R, SQL, Git.
Languages: English (Fluent), Chinese (Native).
Interests: Bass & Electric Guitar, Violin; Photography.